JavaScript Code Explanation Assignment

Objective

Analyze and explain the given JavaScript program through a recorded video explanation. The goal of this assignment is to evaluate your understanding of JavaScript concepts, program flow, asynchronous behavior, validation logic, iterators, and overall code reasoning.

Assignment Instructions

You will be provided with a JavaScript program.

Your task is to:

- Understand the complete code
- Explain the purpose of the program
- Explain how each major function works
- Explain important JavaScript concepts used in the program
- Record your explanation as a video submission.
- **Your video should not exceed 2 minutes.**

What Your Explanation Must Cover

Your video explanation should clearly cover the following areas:

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1. Overall Purpose of the Program

Explain:

- What the application is trying to achieve
- What problem the program solves
- What happens from start to end when the program runs

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2. Program Flow

Explain the execution flow in logical order.

Your explanation should cover:

- Which function runs first
- How data moves through the program
- How applications are stored
- How statuses are updated
- How final output is displayed

The explanation should follow a proper sequence:

Input → Validation → Object Creation → Storage → Status Processing → Final Output

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3. Function-Level Explanation

You should explain the purpose of each important function.

For each function, explain:

- Why the function exists
- What input it receives
- What processing happens inside
- What output/result it produces

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4. Regex and Validation Logic

Explain:

- Why validation is required
- How name validation works
- How email validation works
- What the regular expressions are checking

You should also explain:

- Valid input
- Invalid input
- Error handling behavior

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5. Object Logic

Explain:

- Application object structure
- How application data is stored
- Purpose of idCounter
- How unique IDs are generated

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6. Asynchronous Programming

Explain the asynchronous concepts used in the program.

Your explanation should include:

- Purpose of Promise
- Purpose of setTimeout
- Why async/await is used
- How delayed status updates work

You should explain the real-world reason behind asynchronous behavior in this program.

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7. Custom Iterator

Explain:

- Purpose of Symbol.iterator
- How iteration happens
- What the next() method does
- How applicant names are retrieved one by one

You do not need to memorize definitions, but you should explain the concept clearly in your own words.

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8. Output Explanation

Explain:

- What output is produced
- Why some applications fail

- Why some applications succeed
- How statuses change dynamically

Video Submission Requirements

Duration

Recommended: 2 minutes

Allowed File Formats

Only the following video formats are accepted:

.mp4
.mov
.webm

Evaluation Criteria

Your submission will be evaluated based on:

Criteria	Marks
Regex & Validation Explanation	5
Object Logic	2
Custom Iterator Explanation	4
Asynchronous Programming Explanation	4
Overall Program Summary	3
Flow of Explanation	2

Total: 20 Marks

Important Guidelines

You SHOULD:

- Explain concepts in your own words
- Explain the logic step-by-step

- Use simple language if needed
- Explain the purpose behind the code
- Maintain proper flow in explanation

You SHOULD NOT:

- Read the code line-by-line without explanation
- Skip major functions
- Ignore asynchronous logic
- Ignore iterator concepts
- Give only output without reasoning

Submission Deliverable

Submit:

1 Video File

Accepted formats:

.mp4 / .mov / .webm

Expected Learning Outcome

By completing this assignment, you will be able demonstrate understanding of:

- JavaScript functions
- Regular expressions
- Error handling
- Objects and arrays
- Promises
- Async/Await
- Iterators
- Program flow analysis
- Real-world JavaScript problem solving